MIE312 Midterm 2

The acceleration field

Written by a model for Adit. Not Adit's words.

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1 The acceleration field

Chapter 3 drew a box around a region and balanced what crossed its faces. Chapter 4 shrinks the box to a point, so that the balances hold everywhere in the flow and become equations for the velocity field itself. The first thing Newton's second law needs is an acceleration, and in a fluid that is less obvious than it looks: the velocity is known at fixed points in space, but Newton's law is about a fixed lump of fluid that moves through those points. This section builds the acceleration of a fluid particle out of a velocity field given at fixed points. Everything later in the chapter uses it.

Recognise it. A velocity field given as a formula, $\mathbf{V} = u(x, y, z, t)\,\mathbf{i} + v(\dots)\,\mathbf{j} + w(\dots)\,\mathbf{k}$, and a question asking for "the acceleration" at a point, the largest acceleration in a region, the acceleration along a streamline or centreline, or whether the flow is steady. A nozzle or diffuser modelled by a one-dimensional $u(x)$. Any later question that needs the left side of the Navier–Stokes equation, because that left side is this acceleration times density.

On the crib sheet. Not printed. Section 13 of the sheet gives only the notation for a 2-D field, $\mathbf{V} = u(x, y)\,\mathbf{i} + v(x, y)\,\mathbf{j}$. The material derivative, the split into local and convective parts and the component formulas below have to come from you. They are a chain rule; the model below rebuilds them in under a minute.

The model

Two ways to describe a moving fluid. A *Lagrangian* description follows individual particles: particle number 7 is at position $\mathbf{r}_7(t)$, and its velocity and acceleration are $\dot{\mathbf{r}}_7$ and $\ddot{\mathbf{r}}_7$, exactly as in dynamics. A *Eulerian* description stands at fixed points and records what passes: $\mathbf{V}(x, y, z, t)$ is the velocity of whichever particle happens to be at (x, y, z) at time t. A weather map, a wind-tunnel probe and a CFD result are Eulerian. Fluid mechanics works almost entirely in the Eulerian picture, because tracking every particle is hopeless while a field over space is something you can measure and write down. Here a *field* just means a quantity given as a function of position and time.

The cost is that $\partial\mathbf{V}/\partial t$, the rate of change at a fixed point, is not the particle's acceleration. A fixed point sees a sequence of different particles go past.

The material derivative, by the chain rule. Follow one particle. At time t it is at (x, y, z); a short time dt later it is at $(x+u\,dt,\ y+v\,dt,\ z+w\,dt)$, because it moves with the local velocity. Its velocity changes because time passed and because it moved to a place where the field is different. The chain rule for a function of four variables gives the change:

$$d\mathbf{V} = \frac{\partial \mathbf{V}}{\partial t}\,dt + \frac{\partial \mathbf{V}}{\partial x}\,dx + \frac{\partial \mathbf{V}}{\partial y}\,dy + \frac{\partial \mathbf{V}}{\partial z}\,dz, \qquad dx = u\,dt,\ dy = v\,dt,\ dz = w\,dt.$$

Divide by dt. The acceleration of the particle is

$$\mathbf{a} = \frac{D\mathbf{V}}{Dt} = \underbrace{\frac{\partial \mathbf{V}}{\partial t}}_{\text{local}} + \underbrace{u\frac{\partial \mathbf{V}}{\partial x} + v\frac{\partial \mathbf{V}}{\partial y} + w\frac{\partial \mathbf{V}}{\partial z}}_{\text{convective}} = \frac{\partial \mathbf{V}}{\partial t} + (\mathbf{V}\cdot\nabla)\mathbf{V}. \tag{1}$$

The operator $D/Dt = \partial/\partial t + u\,\partial/\partial x + v\,\partial/\partial y + w\,\partial/\partial z$ is the *material derivative*: the rate of change seen by an observer riding on a particle of material. It applies to any field, not only velocity. DT/Dt is how fast a particle's temperature changes, and $D\rho/Dt$ how fast its density changes. The symbol $\mathbf{V}\cdot\nabla$ is shorthand for $u\,\partial_x + v\,\partial_y + w\,\partial_z$, the rate of change in the direction of motion scaled by the speed.

The two parts have names worth keeping straight.

- *Local* acceleration, $\partial\mathbf{V}/\partial t$: the field itself changing in time at a fixed point. A valve opening, a pump spinning up. It is zero in a *steady* flow, which is a flow whose field does not depend on t.
- *Convective* acceleration, $(\mathbf{V}\cdot\nabla)\mathbf{V}$: the particle being carried ("convected") to a place where the velocity is different. Water speeding up through a steady nozzle has only this kind.

So a steady flow can accelerate. Steady means the picture does not change; it does not mean each particle keeps its velocity. Every particle going through a fixed garden-hose nozzle speeds up.

The components. Written out in Cartesian coordinates, with each line the material derivative of one velocity component:

$$\begin{aligned}
a_x &= \frac{\partial u}{\partial t} + u\frac{\partial u}{\partial x} + v\frac{\partial u}{\partial y} + w\frac{\partial u}{\partial z},\\
a_y &= \frac{\partial v}{\partial t} + u\frac{\partial v}{\partial x} + v\frac{\partial v}{\partial y} + w\frac{\partial v}{\partial z},\\
a_z &= \frac{\partial w}{\partial t} + u\frac{\partial w}{\partial x} + v\frac{\partial w}{\partial y} + w\frac{\partial w}{\partial z}.
\end{aligned}$$

Memory: the component being differentiated stays the same along a line; the multipliers u, v, w go with x, y, z in order.

Along a streamline. A *streamline* is a curve everywhere tangent to the velocity. Measure distance s along it and let V be the speed. In a steady flow the acceleration splits into a part along the streamline and a part toward the centre of its curvature:

$$a_s = V\frac{dV}{ds}, \qquad a_n = \frac{V^2}{R_c},$$

where R_c is the local radius of curvature of the streamline. The first is the convective term written along the path; the second is the centripetal acceleration of circular

motion. On a straight streamline, such as the centreline of a symmetric nozzle, only a_s survives. A particle on a curved streamline at constant speed still accelerates. (With unsteadiness, add $\partial V/\partial t$ to a_s.)

Cylindrical coordinates. With $\mathbf{V} = v_r\,\mathbf{e}_r + v_\theta\,\mathbf{e}_\theta + v_z\,\mathbf{e}_z$, the operator is $D/Dt = \partial_t + v_r\,\partial_r + (v_\theta/r)\,\partial_\theta + v_z\,\partial_z$; the $1/r$ appears because a step $d\theta$ is a distance $r\,d\theta$. The unit vectors $\mathbf{e}_r$ and $\mathbf{e}_\theta$ turn as a particle moves around, and that turning adds two terms:

$$a_r = \frac{Dv_r}{Dt} - \frac{v_\theta^2}{r}, \qquad a_\theta = \frac{Dv_\theta}{Dt} + \frac{v_r v_\theta}{r}, \qquad a_z = \frac{Dv_z}{Dt}.$$

The $-v_\theta^2/r$ is the centripetal acceleration of anything going round. The $v_r v_\theta/r$ is the same term that makes a skater spin faster when pulling in their arms. These reappear in the cylindrical Navier–Stokes equations of the section on the Navier–Stokes equations in the full document.

First moves. Write the three (or two) component lines before substituting anything. Check for t in the field: none means the local term is zero, say so. Compute the nine (or four) partial derivatives as a small table, then substitute the point at the end, not the start. If the question asks for a magnitude, give $|\mathbf{a}| = \sqrt{a_x^2 + a_y^2 + a_z^2}$ and its direction. If it asks for acceleration in g's, divide by $9.81\,\mathrm{m/s^2}$. On a centreline or symmetry line, check whether the cross-velocity is zero there; it usually is, and the problem collapses to $a_s = u\,du/dx$.

Worked examples

Example 1.1 (A squeeze-film press)**.** A flat platen is pressed down onto a thin layer of adhesive spread on a fixed base. The layer is a long strip, so the flow is two-dimensional: x along the strip from its centre, y up from the base. The platen moves down at constant speed $W = 2\,\mathrm{mm/s}$, so the gap is $h(t) = h_0 - Wt$. A plug-flow model of the squeezed-out adhesive is

$$u = \frac{Wx}{h(t)}, \qquad v = -\frac{Wy}{h(t)}.$$

Find the acceleration of the adhesive at $x = 20\,\mathrm{mm}$ at the instant the gap is $h = 0.5\,\mathrm{mm}$.

Assumptions. The given field is taken as exact; it ignores viscosity (it slips at both plates), which is acceptable for a kinematics question. The field depends on t through h, so the flow is unsteady and the local term must be kept.

Check the field first. At $y = 0$, $v = 0$: no flow through the base. At $y = h$, $v = -W$: the fluid moves down with the platen. Good.

Derivatives. Since $dh/dt = -W$, $\partial(1/h)/\partial t = W/h^2$. So

$$\frac{\partial u}{\partial t} = \frac{W^2 x}{h^2}, \quad \frac{\partial u}{\partial x} = \frac{W}{h}, \quad \frac{\partial u}{\partial y} = 0, \qquad \frac{\partial v}{\partial t} = -\frac{W^2 y}{h^2}, \quad \frac{\partial v}{\partial x} = 0, \quad \frac{\partial v}{\partial y} = -\frac{W}{h}.$$

Components.

$$a_x = \frac{W^2 x}{h^2} + \frac{Wx}{h}\cdot\frac{W}{h} + 0 = \frac{2W^2 x}{h^2},$$
$$a_y = -\frac{W^2 y}{h^2} + 0 + \Big(-\frac{Wy}{h}\Big)\Big(-\frac{W}{h}\Big) = 0.$$

Numbers. $a_x = 2(0.002)^2(0.02)/(0.0005)^2 = 0.64\,\mathrm{m/s^2}$, directed outward along the strip, and $a_y = 0$.

Sanity check. In x the local and convective parts are equal and add: the fluid is pushed out faster both because the gap is shrinking (time) and because it moves outward to where the field is faster (space). In y they cancel exactly, which makes sense: a particle at height y keeps the same fraction y/h of the gap, so its vertical velocity is $-Wy/h$ with y/h fixed, a constant. The acceleration grows as $1/h^2$, which is why the last fraction of a millimetre of squeezing is violent.

Example 1.2 (A wind-tunnel contraction, on and off the centreline)**.** A steady model of the flow in a two-dimensional contraction is

$$u = U_0\Big(1 + \frac{x^2}{L^2}\Big), \qquad v = -\frac{2U_0 xy}{L^2},$$

with $U_0 = 5\,\mathrm{m/s}$ and $L = 0.4\,\mathrm{m}$. Find the acceleration (a) on the centreline $y = 0$ at $x = 0.2\,\mathrm{m}$, (b) at $(0.2, 0.1)\,\mathrm{m}$, and (c) at $(0, 0.1)\,\mathrm{m}$.

Assumptions. Steady: no t appears, so $\partial/\partial t = 0$. Two-dimensional: $w = 0$ and nothing depends on z.

Derivatives. $\partial u/\partial x = 2U_0 x/L^2$, $\partial u/\partial y = 0$, $\partial v/\partial x = -2U_0 y/L^2$, $\partial v/\partial y = -2U_0 x/L^2$.

Components.

$$a_x = u\frac{\partial u}{\partial x} + v\frac{\partial u}{\partial y} = U_0\Big(1 + \frac{x^2}{L^2}\Big)\frac{2U_0 x}{L^2} = \frac{2U_0^2 x}{L^2}\Big(1 + \frac{x^2}{L^2}\Big),$$
$$a_y = u\frac{\partial v}{\partial x} + v\frac{\partial v}{\partial y} = -\frac{2U_0^2 y}{L^2}\Big(1 + \frac{x^2}{L^2}\Big) + \frac{4U_0^2 x^2 y}{L^4} = -\frac{2U_0^2 y}{L^2}\Big(1 - \frac{x^2}{L^2}\Big).$$

With $U_0^2/L^2 = 25/0.16 = 156.25\,\mathrm{s^{-2}}$:

(a) $y = 0$: $a_x = 2(156.25)(0.2)(1.25) = 78.1\,\mathrm{m/s^2}$, $a_y = 0$. The centreline is a straight streamline ($v = 0$ there), so this is $a_s = u\,du/dx = (6.25)(6.25) = 78.1\,\mathrm{m/s^2}$, the same.

(b) $a_x = 78.1\,\mathrm{m/s^2}$ (it does not depend on y), $a_y = -2(156.25)(0.1)(0.75) = -23.4\,\mathrm{m/s^2}$, so $|\mathbf{a}| = 81.6\,\mathrm{m/s^2}$, about $8.3g$.

(c) $u = 5\,\mathrm{m/s}$, $v = 0$, $a_x = 0$, but $a_y = -2(156.25)(0.1) = -31.3\,\mathrm{m/s^2}$.

Sanity check. Part (c) is the lesson. At $x = 0$ the particle is moving purely in x at a speed that is momentarily not changing, yet it accelerates toward the centreline: the streamlines converge into the contraction, so they are curved there, and a_y is the centripetal V^2/R_c. Solving $31.25 = 25/R_c$ gives a radius of curvature of $0.8\,\mathrm{m}$ for the streamline through that point. Steady, and accelerating at three g.

Trap. Setting the acceleration to zero because the flow is steady. Steady kills only $\partial\mathbf{V}/\partial t$. The reverse error: treating $\partial u/\partial t$ alone as the acceleration of an unsteady flow. Substituting the point's coordinates before differentiating, which turns every derivative into zero. Dropping the $v\,\partial u/\partial y$ term because "the flow is mainly in x"; it is often the one that matters off the axis. In cylindrical coordinates, forgetting $-v_\theta^2/r$, which is the entire acceleration of a particle in a vortex.

Ladder.

1. White P4.2. A nozzle with linearly rising speed; the acceleration at entrance and exit. One-dimensional convective acceleration.
2. White P4.4. A 2-D converging nozzle field; the accelerations and where the largest is. Both components of the material derivative.

3. White P4.10. A linear plane field; the acceleration and its direction. Reading the acceleration vector geometrically.
4. White P4.8. Unsteady duct flow after a valve opens; local plus convective acceleration. When the local term matters.

Answers are in the back of White; settle each call the same sitting.

Where it hides. The left side of every momentum equation in this chapter is $\rho\, D\mathbf{V}/Dt$, so this section is the first line of the section on the Navier–Stokes equations in the full document, and the exact flows of the section on exact viscous flows in the full document are exactly the cases where it vanishes. The material derivative of density appears in continuity (the section on differential continuity in the full document). The rigid-body line on the crib sheet, $\nabla P = \rho(\mathbf{g} - \mathbf{a})$, is the Navier–Stokes equation for a flow with no shear, with this $\mathbf{a}$. The convective term $u\,\partial u/\partial x$ is what scales as $\rho V^2/L$ in Chapter 5 and puts the density into the Reynolds number. Integrated along a streamline, $a_s = V\,dV/ds$ is where Bernoulli's $V^2/2$ comes from.